

A Weighted Mourre Estimate Without Norm Boundary Values

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Source Question and Answer

Golénia and Jecko derive a limiting absorption principle (LAP) from a strict weighted Mourre estimate by a localised Putnam argument, without differential inequalities. They ask whether this approach gives continuity of the norm boundary values of the sandwiched resolvent [1, pp. 3–4].

LAP (1.2) for H respectively to $(\mathcal{I}', s', A)$, for any $s' \geq s$. Therefore, it is enough to prove Theorem 1.1 for s close to $1/2$.

Remark 1.3. Using Gérard’s energy method in [Gé], we can upper bound the size of the l.h.s. of (1.2) in terms of the constant c_1 appearing in (1.4). See Corollary 3.7

Actually Theorem 1.1 will follow from the more general result obtained in Theorem 3.4. The new theory that we present here and that we call “weighted Mourre theory” is essentially a part of the variant of the Mourre theory in [Gé, GJ]. As such, it is simpler than the usual Mourre theory (it does not use differential inequalities). However, we do not know if such approach gives continuity results on

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the boundary values of the resolvent (1.3). We shall give two (almost equivalent) ways to view the new theory (cf., Subsections 3.2 and 3.3).

As announced above, we want to derive the LAP (1.2) (for some A) on carefully

We give a negative answer at the exact level of their general weighted Mourre theorem: all its strict-estimate, localisation, and conjugation hypotheses can hold while a norm boundary value fails to exist. Thus neither existence nor continuity follows from that estimate and energy argument alone. Status: *candidate full counterexample, likely valid*, subject to human review.

The One-Block Model

On $\mathcal{K} = L^2(\mathbb{R})$ let Q be multiplication by x and $P = -i d/dx$. Set

$$D = (1 + P^2)^{-1}, \quad g(t) = \frac{1}{2} \left(\arctan t + \frac{t}{1 + t^2} \right), \quad b(t) = \frac{\pi}{4} - g(t), \quad B_0 = b(P). \quad (1)$$

The function b is bounded and nonnegative. Since $g'(t) = (1 + t^2)^{-2}$ and $[Q, f(P)] = if'(P)$,

$$[Q, iB_0] = -b'(P) = (1 + P^2)^{-2} = D^2. \quad (2)$$

For $z \notin \mathbb{R}$ write

$$S(z) = D(Q - z)^{-1}D. \quad (3)$$

The positive-commutator energy estimate gives $\sup_{z \notin \mathbb{R}} \|S(z)\| < \infty$; this is also the elementary free one-dimensional LAP with two derivatives of smoothing.

We need two endpoint facts, including the first one in operator norm.

Lemma 1. *There is a nonzero $S_+ \in \mathcal{B}(\mathcal{K})$ such that*

$$\lim_{\tau \downarrow 0} \|S(i\tau) - S_+\| = 0, \quad \|S(iy)\| \leq y^{-1} \quad (y > 0). \quad (4)$$

Proof. For $u = Df$, $v = Dh$, with $f, h \in L^2(\mathbb{R})$,

$$\langle f, S(i\tau)h \rangle = \int_{\mathbb{R}} \frac{\overline{u(x)}v(x)}{x - i\tau} dx. \quad (5)$$

Here $D : L^2(\mathbb{R}) \rightarrow H^2(\mathbb{R})$ is bounded. If $F = \overline{u}v$, then

$$\|F\|_1 + \|F\|_\infty + \text{Lip}(F) \leq C\|f\|_2\|h\|_2 \quad (6)$$

by the one-dimensional Sobolev embedding. Splitting the integrals into $|x| \leq \tau$, $\tau < |x| \leq 1$, and $|x| > 1$ in

$$\frac{1}{x - i\tau} = \frac{x}{x^2 + \tau^2} + i\frac{\tau}{x^2 + \tau^2}$$

gives, uniformly for the functions in (6),

$$\int \frac{F(x)}{x - i\tau} dx = \text{pv} \int \frac{F(x)}{x} dx + i\pi F(0) + O\left(\tau \log \frac{2}{\tau}\right). \quad (7)$$

Thus the right side without the error defines a bounded sesquilinear form, and (7) proves operator-norm convergence to its operator S_+ . This operator is nonzero: choosing $u \in H^2(\mathbb{R})$ with $u(0) \neq 0$ and $f = (1 + P^2)u$ gives

$$\text{Im}\langle f, S_+f \rangle = \pi|u(0)|^2 > 0.$$

Finally, the spectral theorem yields $\|S(iy)\| \leq \|D\|^2\|(Q - iy)^{-1}\| \leq y^{-1}$. $\square$

The Direct-Sum Counterexample

Let $\delta_n = 2^{-n}$ and, on a copy $\mathcal{K}_n$ of $\mathcal{K}$, define

$$H_n = \delta_n Q, \quad B_n = B_0, \quad C_n = \delta_n^{1/2} D.$$

On $\mathcal{H} = \bigoplus_{n \geq 1} \mathcal{K}_n$, take the natural direct sums

$$H = \bigoplus_{n \geq 1} H_n, \quad B = \bigoplus_{n \geq 1} B_n, \quad C = \bigoplus_{n \geq 1} C_n. \quad (8)$$

The operator H is self-adjoint on its direct-sum graph domain, B is bounded self-adjoint and nonnegative, and C is bounded, positive, and injective.

Theorem 1. *The triple (H, B, C) satisfies the strict weighted projected Mourre estimate and every auxiliary hypothesis of [1, Theorem 3.4] on $I = (-2, 2)$ and $I' = [-1, 1]$. Its sandwiched resolvent satisfies the uniform LAP there, but*

$$\lim_{\eta \downarrow 0} C(H - i\eta)^{-1}C \quad \text{does not exist in operator norm.} \quad (9)$$

In particular, the weighted Mourre hypotheses do not imply continuous norm boundary values.

Proof. Each H_n is a rescaled multiplication operator and has no eigenvalues, so the pure-point projection is zero. Equations (2) and (8) give the global form identity

$$[H, iB] = \bigoplus_n \delta_n D^2 = C^2. \quad (10)$$

Consequently the strict weighted projected estimate holds as equality after localisation to any interval.

It remains to check the two conjugations in the abstract theorem. Since B_0 and D are functions of P , CBC^{-1} agrees with B on $\text{Ran } C$ and extends boundedly. Choose $\chi \in C_c^\infty((-2, 2))$ equal to one on $[-1, 1]$. The n th block of $C\chi(H)C^{-1}$ is $D\chi(\delta_n Q)D^{-1}$. For $f_n(x) = \chi(\delta_n x)$, the commutation rule with P gives the exact identity

$$Df_n(Q)D^{-1} = f_n(Q) + 2iDPf'_n(Q) - Df''_n(Q). \quad (11)$$

Because $\|DP\| = 1/2$, $\|D\| = 1$, and $\|f_n^{(k)}\|_\infty \leq \delta_n^k \|\chi^{(k)}\|_\infty$, these blocks are bounded uniformly in n . Thus all hypotheses of the cited theorem hold. They imply the uniform LAP; directly, scaling also gives

$$C_n(H_n - z)^{-1}C_n = S(z/\delta_n). \quad (12)$$

Let $m = \|S_+\| > 0$. From (12),

$$T(\eta) := C(H - i\eta)^{-1}C = \bigoplus_{n \geq 1} S(i\eta/\delta_n). \quad (13)$$

If $T(\eta)$ converged in norm, compression to every fixed block and Lemma 1 would force its limit to be $T_+ = \bigoplus_n S_+$. For every $\eta > 0$, however, one can choose n so large that $y := \eta/\delta_n \geq 2/m$. Lemma 1 then gives

$$\|T(\eta) - T_+\| \geq \|S(iy) - S_+\| \geq m - \|S(iy)\| \geq \frac{m}{2}.$$

This contradicts norm convergence and proves (9). $\square$

Scope, Novelty Check, and Verification

The counterexample satisfies the source's *general* weighted Mourre theorem, from which its principal LAP theorem is derived. It proves that the strict weighted estimate plus the localised Putnam energy argument alone cannot furnish norm boundary values. It does not exclude an additional theorem that uses, simultaneously across all scales, the narrower structure $C = \langle A \rangle^{-s}$ and $B = \varphi(A)$ occurring in Theorem 1.1.

The same continuity limitation was stated explicitly in a 2017 follow-up [2]. A 2020 sequel says that the weighted-Mourre contradiction proof provides no continuity description [3]. Cheap indexes and bounded searches through 17 August 2026 found no later answer. This is not a priority claim.

The companion script verifies the derivative and scaling identities and the dyadic escape mechanism. The proof above is exact; the computation is only an auxiliary transcription check.

Human Review Recommendation

Check the domain-level commutator identity (10), the cutoff conjugation (11), and the uniform form estimate (7). Then audit that fixed-block convergence determines the unique candidate direct-sum limit before the escaping-block lower bound is applied.

References

- [1] S. Golénia and T. Jecko, *Weighted Mourre’s commutator theory, application to Schrödinger operators with oscillating potential*, J. Operator Theory **70** (2013), 109–144; arXiv:1012.0705v3.
- [2] T. Jecko and A. Mbarek, *Limiting absorption principle for Schrödinger operators with oscillating potential*, Documenta Math. **22** (2017), 727–776; arXiv:1610.04369.
- [3] S. Golénia and M.-A. Mandich, *Limiting absorption principle for discrete Schrödinger operators with a Wigner–von Neumann potential and a slowly decaying potential*, arXiv:2002.04909 (2020).